Skills & Personality (SKIPPER) Project

Pilot study: First insights and results

This document summarizes the results using data from the pilot study of the research project Skills & Personality (SKIPPER). It investigates whether individuals can train two of the social, emotional, and behavioral skills (SEB) (Soto et al., 2021). SEB skills are capacities to perform a behavior when it is required by a situation. Overall, five broad skill domains and 32 skill facets are distinguished.

The main objective of this research project is to examine whether emerging adults (18 to 29 years old) can train their social engagement (SE) or emotional resilience (ER) skills through learning. ER skills are capacities to regulate emotions when a situation demands it, while SE skills are capacities to actively engage with other people when a situation calls for it (Soto et al., 2021). ER skills comprise five facets: anger management, impulse regulation, capacity for optimism, stress regulation, confidence regulation. Similarly, SE skills consist of five facets: conversation skills, energy regulation, expressive skill, leadership skill, persuasive skill.

Methods

Design

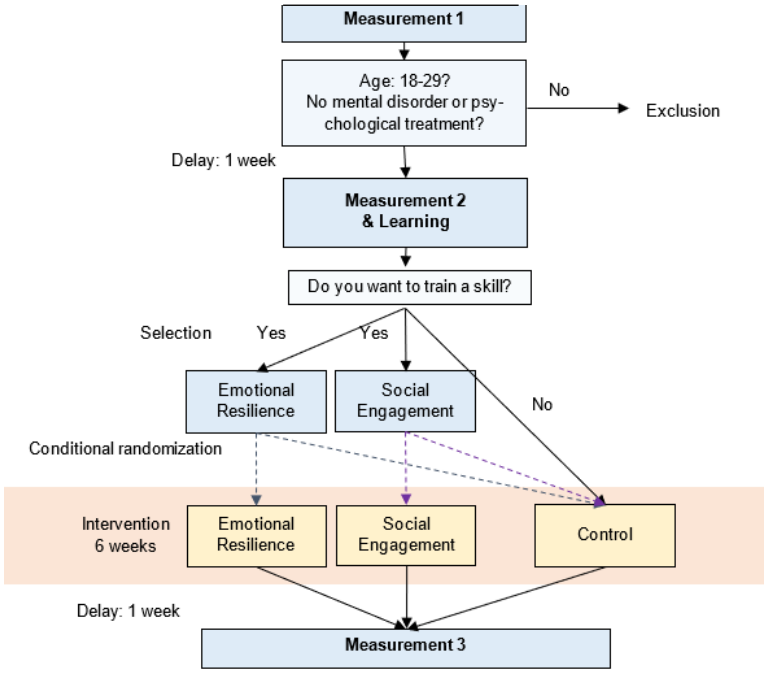
An overview of the study design is shown in Figure 1. First, participants aged 18 to 29 years, without diagnosed psychological disorder or ongoing psychotherapy, completed questionnaires at two measurement points. During the second time point (T2), they learned about personality traits and skills before deciding whether they wanted to train a skill and, if so, which of the two skills to focus on, ER or SE.

Based on this decision, they were randomly assigned to either their preferred training or control group (CG) (i.e., *conditional randomization*). Participants in the intervention groups (IG) trained their selected skills, while those in the CG were only asked about their weekly schedules and to provide a self-reflection of the past week. Note that participants in the IGs were also asked to provide self-reflections of the past week. After six weeks, participants completed questionnaires again (T3).

This project represents a quasi-experimental pre-post control group design, as group assignment is conditional on participants chosen training preference (for more information see: https://osf.io/sfnwt).

Figure 1

Study design of the SKIPPER study project



Sample

Data were collected in Germany. The total sample consisted of N = 102 participants (age: M = 22.48, SD = 3.12, range = 18-29) at the first measurement time point (T1). The majority of the sample was female (85.29%, 11.77% male), and most were either students (80%) or employed (17%). Only n = 3 participants opted not to train a skill (CG). In total, n = 23 of 43 participants who expressed a desire to train their ER skills were assigned in the respective IG, and n = 22 out of 39 participants with the desire to train their SE skills were assigned to the other respective IG. The remaining n = 37 participants were assigned to the CG.

Due to a dropout at the last measurement (T3), the final sample comprised of N = 51 participants (age: M = 22.35, SD = 3.17; 86% female, 7.84% male). In this finisher sample, n = 36 were in the CG, n = 7 in the ER-IG and n = 8 in the SE-IG. Using the available data, we conducted analyses to evaluate the interventions' effectiveness. Please note that we conducted the pilot study to test initial trends in skill change and feasibility of the study. The results, observations, and anonymous feedback from participants were used to make adaptions for the main study.

Measure

All social, emotional, and behavioral skills were assessed with the Behavioral, Emotional, and Social Skills inventory in German language (BESSI-G) (Lechner et al., 2022; Soto et al., 2022). This inventory asks respondents to judge *how well* they can do certain activities described by the items on a scale from 1 = not at all well to 5 = extremely well.

Analysis: evaluation criteria

There are several methods to evaluate intervention effectiveness in a pretest-posttest control group design (PPC). We decided to use three evaluation criteria.

1. Mean-level changes in skills within intervention groups

First, we aimed to detect significant mean-level changes within the IGs in the respective skills applying *t-tests*. For this purpose, each IG was treated as a single-group pretest-posttest design (Morris & DeShon, 2002), ignoring the control group. However, due to the small sample sizes within the IGs, we decided to use effect size statistics to descriptively asses mean-level changes rather than conducting t-tests. We calculated *single-group pretest-posttest change scores* and *raw scores* (Morris & DeShon, 2002, *p.*10). Below, equations for both effect sizes are provided.

The change score d_{change} represents the quotient of the average difference score between two measurement points and sample standard deviation of difference scores as shown in equation 1.0 (Morris & DeShon, 2002, *p.* 10).

$$d_{change} = \frac{M_D}{SD_D} = \frac{M_{post} - M_{pre}}{SD_D} \quad (1.0)$$

Here, the mean score M_{pre} of a construct before the intervention (i.e., T1) is subtracted from the mean score M_{post} after the intervention (i.e., T3). The sample standard deviation of difference scores (SD_D) is calculated as follows:

$$SD_D = \sqrt{\sum_{m=1}^{n} \frac{(d_m - M_D)^2}{n-1}} \quad (1.1)$$

where d_m represents the raw difference scores between the pre and posttest for each participant *m* (cf. the formula for sample standard deviation).

The raw score d_{raw} was calculated by dividing the mean difference score by the standard deviation of the construct measure of the pretest (i.e., before the intervention) (Morris & DeShon, 2002, *p.* 10). This effect size is often used in meta-analyses on mean-level changes (Bleidorn et al., 2022; Roberts et al., 2006; Scherrer & Preckel, 2019).

$$d_{raw} = \frac{M_{post} - M_{pre}}{SD_{pre}} \quad (2.0)$$

We expected to observe at least small mean-level increases of ≥ 0.10 (Cohen, 1988) within each IG. Furthermore, our objective was to demonstrate that more than half of the skill facets within each respective skill domain met this criterion - i.e., three out of five facets within each domain.

2. Meta-analyses: Differences in mean-level changes

Second, we used a meta-analytic approach (Borenstein et al., 2009) to test for significant differences in mean-level changes between the IGs and the CG. Specifically, the coefficients of mean-level changes in facets of a skill domain across all studied groups were used to estimate a summary effect size via a random effects meta-analysis. For example, 15 pretest-posttest raw scores d_{raw} of SE facets were used to estimate a summary mean-level change in SE (i.e., five effect sizes representing the mean-level changes of the five SE facets in the three groups SE-IG, ER-IG, and CG). We chose to use the pretest-posttest raw scores d_{raw}, because these effect sizes are commonly used to estimate the overall mean-level change of constructs in meta-analyses (Bleidorn et al., 2022; Roberts et al., 2006). Moderator analyses were then conducted using *group* as a dummy-coded variable to examine statistical differences between groups (SE-IG: 1 = group is SE-IG, 0 = group is not SE-IG). CG was set as the reference category. The same approach was applied to ER skills.

In addition, we split the CG into two subgroups because participants in this group differed in their preference for a skill training. Therefore, we created a new group variable with SE-CG and ER-CG to distinguish participants who desired to train SE and ER skills, respectively. The same meta-analytic procedure was applied using this variable as a moderator to potentially control for selection effects. The variance of effect sizes was estimated based on an equation proposed by Borenstein and colleagues (2009, *p.* 29). The formula uses the effect size (*d*), the sample size *n*, and the autocorrelation *r* of two measurement points.

$$v_d = (\frac{1}{n} + \frac{d^2}{2n})2(1 - r) \quad (3.0)$$

3. Effect sizes from pretest-posttest-control group design

Third, we quantified the effectiveness of the interventions by calculating an effect size, taking the pretest-posttest-control group design into account (Morris, 2008). The effect size d_{ppc2} was used, which was calculated using the pooled pretest standard deviation in the denominator (*p*. 369):

$$d_{ppc2} = c_p \left[\frac{(M_{post,IG} - M_{pre,IG}) - (M_{post,CG} - M_{pre,CG})}{SD_{pre,pooled}}\right] \quad (4.0)$$

where the indices IG and CG indicate that the means are from the intervention and control groups, respectively. The pooled standard deviation at the pretest is calculated as follows (Morris & DeShon, 2008, *p.* 369):

$$SD_{pre,pooled} = \sqrt{\frac{(n_{IG}-1)SD^2_{pre,IG}+(n_{CG}-1)SD^2_{pre,CG}}{n_{IG}+n_{CG}-2}} \quad (4.1)$$

The c_p within the formula adjusts for biases due to small sample sizes:

$$c_p = 1 - \frac{3}{4(n_T-n_C-2)-1} \quad (4.2)$$

Results

The descriptive statistics for all groups in the pilot study are summarized in Table A in the Appendix.

1. Mean-level changes in skills within intervention groups

The ER skill domain increased within the respective intervention group (ER-IG), showing large effect sizes (d_{change} = 1.13, d_{raw} = 0.65). Furthermore, four out of five ER facets (80%) increased with effect sizes of at least 0.10 (d_{change} = 0.42-0.56, d_{raw} = 0.23-0.68) within this intervention group. Similarly, the SE skill domain increased within the respective intervention group (SE-IG) with d_{change} = 0.20 and d_{raw} = 0.17. Additionally, four out of five SE facets (80%) showed positive pretest-posttest change scores with effect sizes ≥ 0.10 (d_{change} = 0.10-0.45). However, only three out of five SE facets (60%) met the same criterion for d_{raw}. An overview and additional details are displayed in Table 1. The first evaluation criterion was met. The same information for the whole CG, as well as for the separated CGs by training preference (ER-CG, SE-CG), is summarized in Table 1.

2. Meta-analysis: Differences in mean-level changes

Emotional resilience skills. The estimated summary effect for ER skills across the ER-IG, SE-IG, and CG was $\hat{d}_{\text{raw}}$ = 0.16 (*SE* = .04, *CI* = [0.06, 0.25], $p < .01$). In a subsequent meta-regression (moderator analysis), only the SE-IG differed significantly from the CG in their ER facets ($b_{SE\text{-}IG}$ = -0.22, *SE* = 0.11, $p < .05$), while the ER-IG did not differ from the CG ($b_{ER\text{-}IG}$ = 0.05, *SE* = 0.13, $p > .10$). We changed the reference category to test whether the two IGs differed in their ER mean-level change and found that the mean-level change within the SE-IG differed marginally from that in the ER-IG; it was weaker (b_{SE_IG} = -.26, *SE* = .14, $p < .10$). When only ER-IG and the ER-CG groups were investigated the general effect for ER skills was $\hat{d}_{\text{raw}}$ = 0.24 (*SE* = .07, *CI* = [0.12, 0.38], $p < .001$) whereby the mean-level changes did not differ between these two groups (ER-IG vs. ER-CG, $p > .10$).

Social engagement skills. There was no significant increase of SE facets across the IGs and CG ($\hat{d}_{\text{raw}}$ = 0.04, *SE* = 0.04, *CI* = [-0.03, 0.12], $p > 0.10$). We did not find significant differences in mean-level changes between the two IGs compared to the CG. When changing the reference category, no differences were found in the SE changes between the SE-IG and ER-IG. Estimating the overall effect size only for the SE-IG and SE-CG resulted in $\hat{d}_{\text{raw}}$ = 0.01 (*SE* = .05, *CI* = [-0.09, 0.12], $p > 0.10$). No significant differences between these two groups were found ($p > 0.10$). In summary, the second criterion was not fulfilled.

Table 1

Mean-level changes in social, emotional, and behavioral skills

	CG	IG_{ER}	CG_{ER}	IG_{SE-}	CG_{SE}
Emotional Resilience					
d_{change}	0.58	1.13	0.75	-0.08	0.38
d_{raw}	0.33	0.65	0.38	-0.05	0.26
Anger management					
d_{change}	0.42	0.56	0.42	-0.20	0.42
d_{raw}	0.28	0.28	0.19	-0.08	0.49
Confidence regulation					
d_{change}	0.49	<0.00	0.57	0.22	0.44
d_{raw}	0.21	<0.00	0.26	0.19	0.17
Impulse regulation					
d_{change}	0.25	0.54	0.43	-0.44	-0.05
d_{raw}	0.19	0.49	0.32	-0.44	--0.04
Capacity for optimism					
d_{change}	0.40	0.54	0.54	-0.12	0.32
d_{raw}	0.19	0.49	0.28	-0.06	0.13
Stress regulation					
d_{change}	0.40	0.42	0.53	0.35	0.19
d_{raw}	0.25	0.63	0.35	0.16	0.15
Social Engagement					
d_{change}	0.43	-0.08	0.82	0.20	0.05
d_{raw}	0.15	-0.05	0.31	0.17	0.02
Conversation skill					
d_{change}	0.50	-0.22	0.76	0.17	0.34
d_{raw}	0.26	-0.11	0.34	0.13	0.26
Energy regulation					
d_{change}	0.01	.08	0.06	0.14	-0.14
d_{raw}	0.01	0.08	0.05	0.17	-0.06
Expressive skill					
d_{change}	-0.06	0.14	0.22	-0.14	-0.31
d_{raw}	-0.04	0.05	014.	-0.06	-0.20
Leadership skill					
d_{change}	0.17	-0.28	0.32	0.45	>0.00
d_{raw}	0.09	-0.14	0.22	0.36	>0.00
Persuasion skill					
d_{change}	0.39	0.04	0.63	0.10	0.15
d_{raw}	0.19	0.03	0.26	0.05	0.08
Cooperation					
d_{change}	0.04	0.15	0.48	-0.06	-0.40
d_{raw}	0.02	0.06	0.26	-0.04	-0.22
Innovation					
d_{change}	0.01	0.28	0.50	-0.06	-0.64
d_{raw}	0.01	0.19	0.23	-0.04	-0.28
Self-Management					
d_{change}	0.21	0.25	0.34	-0.26	0.03
d_{raw}	0.09	0.11	0.18	-0.10	0.01

Note. CG = control group with n = 36, IG_{ER} = emotional resilience intervention group with n = 7, CG_{ER} = emotional resilience control group with n = 19, IG_{SE-} = social engagement intervention group with n = 8, CG_{SE} = social engagement control group with n = 15; d_{change} = single-group pretest-posttest change score, d_{raw} = single-group pretest-posttest raw score. The results of interest are black.

Additional comparison of the separated control groups. We compared both control groups, separated by their training preference, in their overall mean-level changes. ER skills increased with $\hat{d}_{raw}$ = .18 (*SE* = .05, *CI* = [.08, .28], *p* <.001), with no difference found between ER-CG and SE-CG. The overall mean-level increase in SE skills across both CGs was only marginally significant ($\hat{d}_{raw}$ = 0.09, *SE* = .05, *CI* = [-.01, .19], *p* < .10) Interestingly, SE skills increased positively within the ER-CG, whereas there was no change in the SE-CG. The difference in the SE skill increase between these two groups was significant ($b_{ER\text{-}CG}$ = .20, *SE* = .10, *p* <.05).

3. Effect sizes from pretest-posttest-control group design

Finally, effect sizes accounting directly for the pretest-posttest control group design were calculated (Table 2). The intervention for ER was effective compared to the CG, showing an effect size d_{ppc2} = 0.14. This was also true for five out of five ER facets (80%). In contrast, the intervention for SE was not effective compared to the CG, except for the SE facet *leadership skill*. The picture changes when examining the effect sizes comparing SE-IG with SE-CG. The SE intervention was effective for SE domain as well as for two SE facets: *leadership skill* and *energy regulation*. Overall, the ER intervention met the criterion of an effect ≥ 0.10 when compared with the whole CG. This was not true for the SE intervention.

Table 2

Pretest-posttest control group design effect sizes

	d_{ppc2}				
	IG_{ER}-CG	**IG_{SE}-CG**	**IG_{ER}-CG_{ER}**	**IG_{SE}-CG_{SE}**	**CG_{SE}-CG_{ER}**
Emotional Resilience	0.14	-0.36	0.04	-0.29	-0.20
Anger management	0.10	-0.36	0.15	-0.50	0.11
Confidence regulation	-0.21	<0.00	-0.24	0.03	-0.07
Impulse regulation	0.33	-0.57	0.18	-0.33	-0.36
Capacity for optimism	0.14	-0.25	0.04	-0.21	-0.13
Stress regulation	0.12	-0.07	-0.02	0.08	-0.31
Social Engagement	-0.19	-0.02	-0.32	0.11	-0.27
Conversation skill	-0.36	-0.15	-0.44	-0.11	-0.13
Energy regulation	0.03	0.09	<0.00	0.16	-0.11
Expressive skill	0.09	-0.04	-0.06	0.09	-0.32
Leadership skill	-0.23	0.23	-0.33	0.30	-0.17
Persuasion skill	-0.14	-0.12	-0.24	-0.12	-0.21
Cooperation	0.07	-0.07	-0.15	0.15	-0.47
Innovation	0.23	-0.05	0.04	0.23	-0.50
Self-Management	0.02	-0.19	-0.05	-0.10	-0.15

Note. d_{ppc2} = effect sizes taking the pretest-posttest control group design into account (Morris, 2008), CG =control group with n = 37, IGER = emotional resilience intervention group with n = 7, IGSE-= social engagement intervention group, CGER = control group with n = 19 participants desired to train emotional resilience, CGSE = control group with n = 15 participants desired to train social engagement.

4. Exploratory analyses: Reliable change index

Given the small sample sizes within the studies, we calculated individual reliable change indices (RCIs) for each participant. We used the test-retest reliability coefficients from Lechner and colleagues (2022), and the standard deviation of the entire sample at T1 of this pilot study. The percentages of participants with RCIs ≥ 1.50 and ≥ 1.96 are presented in Table 3. Descriptively, only small differences were observed.

Table 3

Percentual quantity of participants with reliable change indices ≥ 1.5 and ≥ 1.96

	CG		IG_{ER}		CG_{ER}		IG_{SE-}		CG_{SE}	
Reliable change index ≥	**1.5**	**1.96**	**1.5**	**1.96**	**1.5**	**1.96**	**1.5**	**1.96**	**1.5**	**1.96**
Emotional Resilience	25%	11%	29%	0%	32%	16%	0%	0%	20%	7%
Anger management	19%	8%	43%	0%	16%	11%	13%	0%	27%	7%
Confidence regulation	0%	0%	14%	14%	16%	0%	13%	13%	0%	0%
Impulse regulation	11%	3%	43%	0%	0%	5%	0%	0%	7%	7%
Capacity for optimism	17%	14%	29%	29%	26%	21%	13%	0%	7%	0%
Stress regulation	8%	3%	0%	0%	16%	5%	0%	0%	0%	0%
Social Engagement	8%	3%	14%	0%	11%	5%	25%	13%	7%	0%
Conversation skill	22%	8%	0%	0%	26%	0%	13%	0%	20%	13%
Energy regulation	5%	0%	0%	0%	11%	0%	0%	0%	0%	0%
Expressive skill	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Leadership skill	8%	6%	0%	0%	11%	5%	25%	25%	7%	7%
Persuasion skill	14%	6%	14%	14%	21%	11%	13%	0%	7%	0%
Cooperation	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Innovation	0%	0%	14%	0%	0%	0%	0%	0%	0%	0%
Self-Management	3%	0%	0%	0%	5%	0%	0%	0%	0%	0%

Note. CG =control group with n = 37, IG_{ER} = emotional resilience intervention group with n = 7, IG_{SE-}= social engagement intervention group, CG_{ER} = control group with n = 19 participants desired to train emotional resilience, CG_{SE} = control group with n = 15 participants desired to train social engagement.

Summary

The pilot study demonstrates that mean levels of the ER and SE skills changes positively within the IGs with at least a small effect size. This was true for at least 60% of the skill facets within each respective skill domain. However, when using the facet's mean-level changes as effect sizes in meta-analyses, no significant differences were found between the IGs and the CG. We hypothesize that the variability between facets makes the use of their mean-level changes as separate effect sizes less appropriate for representing the broader skill domain in meta-analyses. Furthermore, the ER intervention was effective compared to the entire CG when effect sizes accounting for the pretest-posttest control group design were calculated. The SE intervention was not. In summary, while some positive trends were observed, the small sample sizes in the IGs limit the generalizability of the findings. Thus, the descriptive results reported here may be by chance. Further research with a greater sample is needed.

For the main study, the design was modified to reduce participant burden and improve feasibility (reasonability, economy). Changes included reducing the number of measurement occasions to a single pre-intervention assessment and reduction the number of questionnaires (see https://osf.io/sfp9u). Additionally, the intervention materials for the social engagement

group were adapted based on anonymous participant feedback. Finally, we planned a follow-up assessment six months later to evaluate lasting or delayed effects.

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Table A

Descriptive statistics across all groups of the SKIPPER-pilot study

	CG		ER-IG		ER-CG		SE-IG		SE-CG	
	T1	T3	T1	T3	T1	T3	T1	T3	T1	T3
	M (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)	*M* (*SD*)
Emotional Resilience	3.03 (0.47)	3.19 (0.52)	2.76 (0.34)	2.98 (0.43)	3.02 (0.52)	3.22 (0.52)	2.92 (0.62)	2.89 (0.52)	3.09 (0.40)	3.19 (0.51)
Anger management	3.24 (0.72)	3.44 (0.71)	2.88 (1.02)	3.17 (0.58)	3.46 (0.89)	3.46 (0.81)	3.19 (1.00)	3.10 (0.90)	3.20 (0.52)	3.46 (0.64)
Confidence regulation	3.11 (0.72)	3.26 (0.79)	3.07 (0.63)	3.07 (0.97)	3.28 (0.68)	3.46 (0.74)	2.85 (0.77)	3.00 (0.53)	2.99 (0.72)	3.11 (0.75)
Impulse regulation	3.00 (0.67)	3.13 (0.69)	2.83 (0.73)	3.19 (0.75)	2.93 (0.72)	3.16 (0.73)	3.02 (0.57)	2.77 (0.53)	3.12 (0.64)	3.10 (0.70)
Capacity for optimism	3.03 (0.74)	3.17 (0.70)	2.64 (0.49)	2.88 (0.68)	3.03 (0.72)	3.23 (0.64)	2.85 (0.94)	2.79 (0.55)	3.07 (0.79)	3.13 (0.54)
Stress regulation	2.77 (0.74)	2.92 (0.70)	2.38 (0.34)	2.60 (0.58)	2.55 (0.64)	2.78 (0.60)	2.69 (0.63)	2.79 (0.52)	3.09 (0.40)	3.19 (0.51)
Social Engagement	3.11 (0.51)	3.19 (0.57)	3.30 (0.35)	3.28 (0.43)	3.29 (0.46)	3.43 (0.47)	2.84 (0.39)	2.91 (0.53)	2.97 (0.50)	2.98 (0.55)
Conversation skill	3.00 (0.79)	3.20 (0.80)	3.50 (0.89)	3.40 (0.63)	3.30 (0.78)	3.56 (0.62)	2.92 (0.62)	3.00 (0.56)	2.73 (0.65)	2.90 (0.77)
Energy regulation	3.19 (0.62)	3.19 (0.65)	3.29 (0.30)	3.31 (0.41)	3.19 (0.53)	3.22 (0.64)	3.02 (0.37)	3.08 (0.47)	3.29 (0.72)	3.24 (0.70)
Expressive skill	3.19 (0.73)	3.17 (0.89)	3.62 (0.91)	3.67 (0.90)	3.36 (0.71)	3.46 (0.86)	2.98 (1.06)	2.92 (0.81)	3.13 (0.67)	3.00 (0.73)
Leadership skill	3.22 (0.66)	3.28 (0.62)	3.21 (0.70)	3.12 (0.62)	3.46 (0.47)	3.56 (0.51)	2.54 (0.58)	2.75 (0.66)	2.98 (0.77)	2.98 (0.61)
Persuasion skill	2.97 (0.81)	3.12 (0.80)	2.88 (0.77)	2.90 (0.89)	3.12 (0.89)	3.35 (0.80)	2.75 (0.92)	2.79 (1.03)	2.73 (0.67)	2.79 (0.73)
Cooperation	3.73 (0.36)	3.74 (0.41)	3.50 (0.62)	3.54 (0.48)	3.78 (0.37)	3.87 (0.45)	3.52 (0.53)	3.50 (0.45)	3.72 (0.34)	3.64 (0.28)
Innovation	3.59 (0.41)	3.60 (0.42)	3.40 (0.54)	3.50 (0.23)	3.60 (0.38)	3.69 (0.40)	3.02 (0.48)	3.00 (0.61)	3.67 (0.42)	3.56 (0.43)
Self-Management	3.65 (0.51)	3.69 (0.51)	3.88 (0.52)	3.94 (0.63)	3.65 (0.48)	3.74 (0.52)	3.61 (0.51)	3.56 (0.52)	3.67 (0.59)	3.68 (0.53)

Note. CG =control group with n = 37, IGER = emotional resilience intervention group with n = 7, IGSE-= social engagement intervention group, CGER = control group with n = 19 participants desired to train emotional resilience, CGSE = control group with n = 15 participants desired to train social engagement.